Neural Decoder for Monkey Hand Trajectory Estimation

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***Abstract*— This work addresses the problem of neural decoding of hand trajectories from population spike train data. We propose a causal decoding framework that estimates the two-dimensional hand position from neural activity recorded during reaching movements. The neural signals are first transformed into causal features using binned spike counts over a short temporal history window. To reduce dimensionality and noise, principal component analysis (PCA) is applied to the feature space. A ridge regression model is then trained to map the reduced features to hand positions. Additionally, a direction-based strategy is employed, where separate decoders are trained for each movement direction and selected during inference based on neural activity patterns. The proposed method achieves a root mean squared error (RMSE) of 14.35 cm on the validation data, demonstrating stable and reliable decoding performance.**

I. INTRODUCTION

Brain–machine interfaces (BMIs) aim to translate neural activity into control signals for external devices, with important applications in neuroprosthetics and assistive technologies. A core problem in this area is neural decoding, where recorded neural signals must be used to estimate intended movement.

The present task considers decoding the two-dimensional hand trajectory of a monkey during a reaching experiment. The dataset contains spike train recordings from 98 neural units collected while the monkey repeatedly reached towards targets in eight different directions. For each trial, neural activity is recorded at 1 ms resolution together with the corresponding hand position trajectory, and the objective is to estimate the hand position in the horizontal plane over time from neural activity alone. Since the competition requires causal decoding, only past and present neural information can be used at each time point.

This problem is challenging for several reasons. First, neural spike trains are sparse and noisy. Second, the data are high-dimensional, as activity from many neural units must be integrated over time. Third, the relationship between neural activity and movement is dynamic, meaning that both movement direction and fine-grained trajectory information must be captured by the decoder.

A simple baseline pipeline was first implemented using direction templates for coarse classification, followed by principal component analysis (PCA) and ridge regression for trajectory prediction. This baseline was computationally efficient, easy to interpret, and suitable as an initial reference method. An improved competition pipeline was later developed, in which direction classification was refined using Linear Discriminant Analysis (LDA) and trajectory prediction was performed using Principal Component Regression (PCR). Although both methods are based on relatively lightweight linear models, the latter provides a more structured treatment of classification and regression and achieved stronger decoding performance.

Comparison between the baseline pipeline and the improved competition pipeline provides insight into how different design choices affect decoding accuracy, robustness, and interpretability.

II. METHOD

A. Per-direction, Per-time PCA + Ridge Decoder

The first pipeline used a causal PCA-ridge decoder with separate models for movement direction and decoding time. Spike trains were binned in 20 ms windows, the latest K = 5 bins were concatenated into a causal history feature, and all aligned training samples were pooled to learn a single global PCA projection. Ridge regressors were then trained in the reduced space to estimate the two-dimensional hand position.

For a decoding time t, spike counts from the latest K = 5 bins (bin size 20 ms) were concatenated to form a causal feature vector:

$$\phi(t) = [c_{t-K+1}^T, c_{t-K+2}^T, \ldots, c_t^T]^T \tag{1}$$

All aligned training features from all directions and time points were pooled to compute a single PCA projection:

$$z(t) = U^T \left(\phi(t) - \mu\right) \tag{2}$$

A separate weight matrix $W(d, i)$ is trained for each movement direction d and decoding time index t_i. During inference, direction is first estimated using cumulative spike counts up to $t_{start} = 320$ ms. The corresponding PCA projection is then applied, and the position is estimated as:

$$\hat{p}(t) = W(d, i) \begin{bmatrix} z(t) \\ 1 \end{bmatrix} \tag{3}$$

The implemented setting used $\lambda = 100$ for ridge regression, while the intercept term was left unregularized. This design kept the decoder causal at all stages and allowed a separate mapping to be used for different reach directions and trajectory stages.

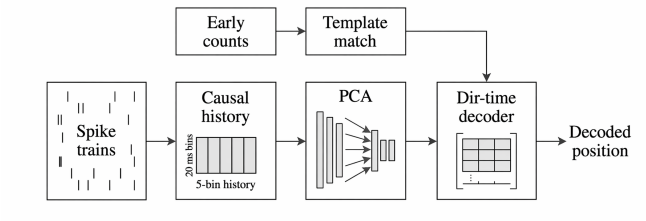

Fig. 1. Simplified overview of the per-direction, per-time PCA + ridge decoder.

B. Improved Method

In this pipeline, four stages are included. The first stage is preprocessing, the second is dimensionality reduction, the third is direction classification, and the fourth is position estimation.

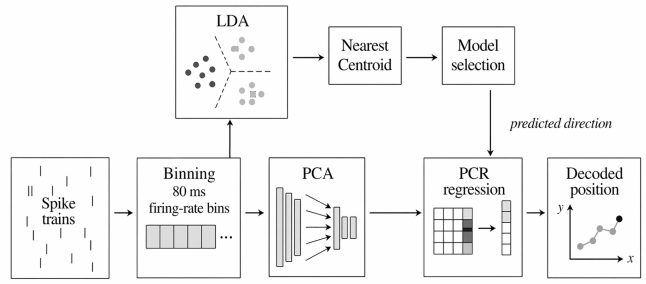

Fig. 2. Overview of the improved decoding pipeline.

For preprocessing, the discrete spikes from the training data are converted into a continuous numerical input (Binning) by averaging the firing rate over a certain period (Δt; in this case, 80 ms). This preprocessing smooths the data and provides a richer gradient for pattern discovery, helping eliminate jitter in each spike and preserving the overall changes among spikes related to the arm's movement. Hence, the firing rates provide a stronger statistical signal showing whether the neurons are involved in the specific arm's movement.

The equation is shown below:

$$V_i = \frac{1}{\Delta t} \sum_{t=t_1}^{t_2} S_i(t) \tag{4}$$

Where $S_i(t)$ is the binary spike data (0 or 1) at time t, and Δt is the duration of the bin $(t_2 - t_1)$.

In the dimensionality reduction stage, Principal Component Analysis (PCA) is used to extract the strongest features from the bins calculated above, discard the remaining data, and reduce processing time. In this case, a fixed number of feature dimensions is used to ensure the same number of components are passed to the Linear Discriminant Analysis (LDA) for classification.

The equation of PCA is shown below:

$$Z_{pca} = W_{pca}^T (f - \mu) \tag{5}$$

Where W_{pca}^T is the matrix of the top fixed number of eigenvectors of the covariance matrix, and μ is the mean feature vector.

Afterwards, to achieve a smaller root mean squared error (RMSE), the firing rate bins are classified into 8 directions corresponding to their highest firing rate, so that each direction will have a specific weighted method for prediction instead of using a single general method for all 8 directions. Here, LDA and Nearest Centroid (NC) methods are used.

During LDA, the PCA components are projected into a certain dimensional space to maximise the spacing between the data bins, forming 8 clusters. According to the results, 6-dimensional is used in this case for the smallest RMSE.

To find the maximum spacing for clear classification, the maximum ratio of between-class scatter (S_B) to within-class scatter (S_W) is calculated based on the following equation:

$$J(W) = \frac{|W^T S_B W|}{|W^T S_W W|} \tag{6}$$

For the direction classification stage, once the bin data are separated into 8 clusters, the mean value of each direction in LDA is found as the centre (centroid). NC is then used to measure the distance between those centres and the bin data, and classify it as the direction whose centre is closest. Thus, even after the new data is received, NC can still determine the direction of the trial data and classify all the trials. The equation of NC is shown below:

$$\hat{c} = \arg\min_k \sum_{j=1}^{M_{LDA}} (Z_j - \bar{Z}_k)^2 \tag{7}$$

Where Z is the current LDA-projected vector, $\bar{Z}_k$ is the mean vector for direction k, and $\hat{c}$ is the predicted direction.

Finally, for position estimation, the classified data will be utilised by Principal Component Regression (PCR), a linear regression model, to predict the arm's movement coordinates. Each direction has its PCR model with specific weights based on the training data. The direction-classified trial is first transformed into the number of trials minus 1 $(N - 1)$ principal components, which can prevent singularities, uncouple the neurons for calculations, and prevent overfitting by focusing on the strongest principal components.

The principal components are then mapped linearly by the weight matrix to find the x and y coordinates. The equations are as shown below:

$$x(t) = f_{reg}^T \beta_x + \alpha_x \tag{8}$$

$$y(t) = f_{reg}^T \beta_y + \alpha_y \tag{9}$$

Where f_{reg} is the vector of neural firing rates, β is the vector of regression coefficients, and α is the bias.

C. Results and Discussion

Maybe choose one main method and discuss its result. then have a discussion of the improvements– junkai

References